too to think of the gene complex as being divided up into discrete replicators or genes. This arises because of the phenomenon of sex. Sexual reproduction has the effect of mixing and shuffling genes. This means that any one individual body is just a temporary vehicle for a short-lived combination of genes. The *combination* of genes that is any one individual may be short-lived, but the genes themselves are potentially very long-lived. Their paths constantly cross and recross down the generations. One gene may be regarded as a unit that survives through a large number of successive individual bodies. This is the central argument that will be developed in this chapter. It is an argument that some of my most respected colleagues obstinately refuse to agree with, so you must forgive me if I seem to labour it! First I must briefly explain the facts of sex.

I said that the plans for building a human body are spelt out in 46 volumes. In fact this was an over-simplification. The truth is rather bizarre. The 46 chromosomes consist of 23 *pairs* of chromosomes. We might say that filed away in the nucleus of every cell are two alternative sets of 23 volumes of plans. Call them Volume 1a and Volume 1b, Volume 2a and Volume 2b, etc., down to Volume 23a and Volume 23b. Of course the identifying numbers I use for volumes and, later, pages, are purely arbitrary.

We receive each chromosome intact from one of our two parents, in whose testis or ovary it was assembled. Volumes 1a, 2a, 3a, … came, say, from the father. Volumes 1b, 2b, 3b, … came from the mother. It is very difficult in practice, but in theory you could look with a microscope at the 46 chromosomes in any one of your cells, and pick out the 23 that came from your father and the 23 that came from your mother.

The paired chromosomes do not spend all their lives physically in contact with each other, or even near each other. In what sense then are they ‘paired’? In the sense that each volume coming originally from the father can be regarded, page for page, as a direct alternative to one particular volume coming originally from the mother. For instance, Page 6 of Volume 13a and Page 6 of Volume 13b might both be ‘about’ eye colour; perhaps one says ‘blue’ while the other says ‘brown’.

Sometimes the two alternative pages are identical, but in other cases, as in our example of eye colour, they differ. If they make contradictory ‘recommendations’, what does the body do? The answer varies. Sometimes one reading prevails over the other. In the eye colour